

METALANGUAGE: THE TECHNOLOGICAL FRAMEWORK

In the sphere of scientific activity there are profound affinities between language and method. Both are ways of constituting specific fields of investigative and controlling action (Tseëlon, 1991). If it is to qualify as scientific, method must be explicitly described, and this cannot be done without reference to the kinds of things the method is applied to. A scientific method is never just a wordless or signless practice; it is a practice under some description. So the practitioners' understanding of what they are doing is intimately tied up with their understanding of the world in which or to which they are doing it. On the other hand, a language is never just a system of representation; as soon as it is used it becomes a practice. People 'do things with words' (Austin, 1962); in speaking or writing they carry out performances that have significance for their world. So language and method are intimately bound to each other.

That bond may become manifest in a variety of forms. The links between language and method may vary in closeness, and there may be large differences in the relative importance of one or other pole in this relationship. In this chapter I want to explore the historically unique constitution of the language–method bond that occurred during a crucial period in the development of modern Psychology. That involves tracing the most significant steps by which the language of psychological methodology came to provide an overarching categorical framework for representing psychological reality.

Like biology, methodology was the source of a metalanguage applied to already existing psychological categories. A key term in this metalanguage was 'variable'. When they deployed the methodological metalanguage, psychologists were no longer looking at slices of reality called 'motives', 'attitudes' or 'personality' – they were studying 'motivational variables', 'attitudinal variables' or 'personality variables'. The new metalanguage did not replace the older biological framework; it merely added an extra layer of description. That layer, however, came to play a more thoroughly hegemonic role than the older framework ever did. By the 1950s use of the methodological metalanguage had virtually become mandatory in the context of psychological investigation and theorizing.

The significance of stimulus-response psychology

Talk of 'variables', however, represented a later phase in the development of a method-based metalanguage. Some time earlier, the language of stimulus and response had come to assume a similar role. It has often been observed that the agenda of behaviourism essentially amounted to a proposal for translating all the mentalistic concepts of psychology into a physicalistic language. But that was not the only change in psychological language that was associated with the earlier forms of behaviourism. There was also a strong preference for describing all psychological processes in stimulus-response terms. In fact, this second item on the behaviourist agenda was initially more acceptable than the first. The majority of American psychologists were not particularly sympathetic to J.B. Watson's crude and extreme physicalism (Samelson, 1981), but stimulus-response psychology seemed an altogether more reasonable idea, at least in experimental circles. That was because stimulus-response psychology seemed to make practical sense. There were in fact two rather different reasons for advocating stimulus-response psychology in the 1920s: because of a preference for a reductionistic psychology based on a mechanistic biology; and because of a desire to treat the standard method of psychological experimentation as the bedrock that would provide a firm foundation for psychological concepts. If one gave pride of place to the second reason, one could easily be a stimulus-response psychologist without being a card-carrying behaviourist.

The prime example for this second option is the immensely influential Columbia psychologist, R.S. Woodworth, who explicitly advocated stimulus-response psychology, but who, when he asked himself whether he was a behaviourist, essentially decided that he didn't know and didn't much care (Woodworth, 1939). Woodworth's position certainly represented mainstream American experimental psychology much more closely than did Watson's. His general textbook bore the title *Psychology: A Study of Mental Life* when it first appeared in 1921, its subtitle a real snub for the behaviourists. But it became by far the most widely used text of its time (Winston, 1988) and saw several subsequent editions. It was followed in 1938 by the authoritative graduate manual *Experimental Psychology*, popularly known as the 'Columbia Bible'.

Although he was a very successful textbook author, Woodworth was first and foremost an experimentalist, actively engaged in laboratory work throughout his career. He always regarded the laboratory as the fountainhead of solid psychological knowledge, and his advocacy of stimulus-response psychology was closely connected with that conviction. It was because psychological experimentation was essentially concerned with stimulus-response relations that the schema of stimulus and response could and should function as the basis for psychological explanation.

Whenever we have any human action before us for explanation, we have to ask what the stimulus is that arouses the individual to activity, and how he responds. Stimulus-response psychology is solid, and practical as well; for if it can establish the laws of reaction, so as to predict what responses will be made to a given stimulus, and what stimulus can be depended on to arouse a desired response, it furnishes the 'knowledge that is power'. Perhaps no more suitable motto could be inscribed over the door of a psychological laboratory than these two words, 'Stimulus-Response'. (Woodworth, 1921: 68)

In advocating the practicality of stimulus-response Woodworth was certainly not oblivious to the possibilities of practical psychological applications for purposes of social control. In fact, at the very beginning of his book he tried to sell the discipline of Psychology on the basis of its supposed military usefulness during World War I. But closer attention to the textual context and to his less popular writings shows quite clearly that it was the practicality of the laboratory that was closest to his heart. He immediately proceeds to justify the above passage by pointing out that even 'the modern introspectionist' would 'find a congenial home in a stimulus-response laboratory'. And a few years later he would claim that, while he had no quarrel with 'the concrete experiments' of Gestalt psychology, the empirical facts fitted a linear stimulus-response account much better than they did the Gestalt 'system' (Woodworth, 1927).

Like most American psychologists at the time, Woodworth was preoccupied by and unhappy about the division of Psychology into 'schools'. There seems to have been a strong feeling that this division undermined the discipline's claims to scientific status. Science was supposed to be the source of certain knowledge which was grounded in the laboratory, rather than in mere theory. So if psychologists would just agree to play down their theoretical differences and concentrate on the experimental procedures which they had in common, they would collectively benefit. Like so many of his colleagues, Woodworth was much concerned with what he called 'the solidarity of the psychological group' (1930: 327), and he thought that this could best be achieved 'from below upwards', by appealing to 'the actual research of psychologists' (*ibid.*), rather than their theoretical predilections. That meant finding the lowest common denominator of experimental practice (which he identified with science in general) and building consensus on that. Now, this basic feature seemed to him to lie in the fact that science 'always deals with systems that are subject to outside influences' (1930: 329). Science devoted itself to the study of these outside influences, which, in Psychology, took the form of stimulus-response relationships.

Woodworth strongly repudiated the suggestion that his proposal did no more than add another variant to the list of psychological schools. Although he never referred to stimulus-response psychology as a language, he used it as such. It provided him with the means for redescribing all substantive psychological terms and categories in a way

that immediately characterized them as topics for experimental investigation. Confronted with any psychological phenomenon whatever, the psychologist is to ask, 'what is the stimulus, and what the response?' (Woodworth, 1921: 484). Woodworth shows how this rule can be applied even to such seemingly recalcitrant topics as play, dreams, thought and imagination. Whatever else might be said about these and other psychological phenomena, they must be translated into stimulus-response terms when they enter scientific psychological discourse. One very important effect of this translation is that it gives exemplary, paradigmatic, status to psychological experiments in interpreting psychological events in general. In this way, experiments on delayed reactions in animals can become paradigmatic for the understanding of human purpose, experiments on paired associate learning can elucidate the nature of memory, experiments on muscular fatigue can provide the model for understanding any acquired predisposition, and reaction time experiments can become the basic exemplar for all behaviour. Invariably, translation into the metalanguage of stimulus and response is required for any of these analogies to work.

In the two decades after the first appearance of Woodworth's basic text this language became pervasive in the psychological literature. The committed behaviourists were not likely to find fault with it, but many more American psychologists probably shared Woodworth's profound indifference to the niceties of psychological theory as well as his equally profound respect for experimental methodology. He had indeed been quite sensitive to a fundamental feature of that methodology, as it had come to be used in psychological laboratories. Quite generally, psychological experiments involved exposing their subjects to a specific set of varied external conditions and recording specific aspects of their conduct under these conditions. Broadly speaking, psychological experiments did have a stimulus-response structure, or at least, that was how they were typically described in the psychological literature. Therefore, description of objects of psychological interest in terms of the stimulus-response language always carried the clear implication that the fundamental structure of these objects was entirely congruent with the structure of psychological experiments. As already indicated, this entailed the important corollaries that (a) these objects of psychological interest could most appropriately be investigated in the laboratory, and conversely, (b) that such laboratory investigations were of exemplary significance for the understanding of the objects of interest. It is not surprising that stimulus-response terminology became a popular *lingua franca* among experimentally oriented psychologists.

Two fundamental features of this terminology have had long term consequences for the discipline. In the first place, psychologists' understanding of their own practice was affected. The language of stimulus and response is an asocial and impersonal language that requires translation of all terms referring to social or specifically human

relationships into a terminology that makes no categorical distinction between these and essentially biological categories. This was touched on in previous chapters. Insofar as discourse about psychological practice was couched exclusively in stimulus-response language, blindness to the intrinsically social nature of that practice prevailed. Psychological experiments involving human subjects are of course social situations that require the highly patterned interaction of the participants (Danziger, 1990a). Without the experimenter's authority (an intrinsically social concept), or without verbally transmitted understandings among human agents, there would be no experiment. But in the language of stimulus and response such features became invisible. All that was supposedly going on was that the experimental subject reacted to stimuli. In this way the fiction that there was no essential difference between psychological experiments and experiments in the natural sciences could be maintained and the occasional voice that questioned this (Rosenzweig, 1933) could be ignored. It was not until the 1960s that there was any kind of discussion of the social features of psychological experimentation, but by then the weight of traditional conceptions was too strong to permit any widespread confrontation of the underlying issues (Gadlin and Ingle, 1975).

A second long term consequence involves psychologists' understanding of the relationship between their empirical practice and their theoretical concepts. A fundamental change in that understanding became manifest once American experimentalists had discarded the pre-World War I influence of earlier European models. In the era of Wundt, when psychology was mentalistic, there was an inevitable disjunction between the practice of experimental psychology and the theoretical constructions used to explain the outcome of experiments. Experimental practice was based on something like a stimulus-response schema. Stimuli, like visual distances or auditory intervals, would be presented and then some kind of reaction, a sensory judgement or motor response perhaps, would be obtained from the experimental subject. But the theoretical processes invoked to explain the results of such experiments had an altogether different structure. Here we find concepts like unconscious inference, apperception or creative synthesis which represent integrative cognitive events that transcend the stimulus-response manipulations characteristic of experimental methodology. By 1920, however, such theoretical concepts had been declared unscientific and redundant. In future, all theoretical concepts would have to be translated into stimulus-response terms if they were to pass muster. In other words, the specially created structure of events in the psychological laboratory came to function as a kind of model or template for the theoretical construction of the objects of psychological investigation.

Not all areas of science are characterized by the same relationship between theory and experimental practice (Knorr Cetina, 1992). For example, experiments can be regarded as yielding *signs* that point to an

underlying reality, or they can be regarded as an actual *simulation* of the part of reality being investigated. In the history of experimental psychology there seems to have been a change from the first to the second of these conceptions, a change that coincided with the shift of the discipline's centre from Germany to America. Initially, experimental psychology had inherited its object of investigation, the typical adult consciousness, from a particular cultural and philosophical tradition. It merely applied special techniques to the study of this object. But once the disciplinary apparatus of investigation had been institutionalized the possibility emerged of allowing this apparatus, rather than tradition, to define the objects of psychological science. The realization of this possibility was inseparably bound up with an increase in the social influence of the practitioners of psychological science. It was *their* practice which was to play the constituting role in defining the structure of psychological reality. In this sense the discourse of psychology became a discourse of practice, the practice of the experts.

But that development would not have been as successful as it was if the new language of psychology had not had a certain appeal to social interests outside the discipline. By limiting itself to a language that directly reflected the practical manipulations of experts the new psychology pandered to dreams of 'social control' within which most social problems were transformed into individuals' problems that could be solved by applying the appropriate technological fix. For the new psychological discourse that spread so rapidly in North America during the years following World War I psychological reality was defined as the sum of the possibilities of expert psychological manipulation.

Within the discipline, the popularity of the new conception of the relation between experimental and theoretical reality undoubtedly owed something to its compatibility with mechanistic concepts of the reflex that had come down from the nineteenth century. But although the biological background of stimulus-response terminology might have helped to establish it in the first instance, there was always a significant minority of more clinically, or more humanistically, oriented psychologists who were not impressed by this background. As a neutral language that all psychologists might feel comfortable using, stimulus-response had its defects. Starting in the 1930s, we find its place gradually being taken by a substitute that preserved the basic conception of the relationship between experiment and theory but that avoided the mechanistic echoes from the past. This substitute was the language of variables.

Psychologists encounter 'variables'

By the 1950s, most psychologists had adopted the habit of referring to any of the things they investigated as 'variables'. Whether it was a

matter of specifying features of the social or physical environment, or a matter of categorizing dispositions, actions or attributes of individuals, psychological theory and research was apt to define them all as 'variables'. This is quite remarkable, in view of the fact that modern psychology flourished for more than half a century before much was heard of 'variables'. There are rare instances where an early psychologist uses the term 'independent variable',¹ but there is not even a hint of widespread or systematic use before the 1930s. As Winston (1988, 1990) has noted, the term begins to crop up at that time in the writings of prominent figures like Tolman (1932), Woodworth (1934) and Boring (1933).

For Woodworth and other stimulus-response psychologists the transition to the language of variables was quite simple. What had previously been referred to as a stimulus was now categorized as an 'independent variable'; similarly, responses became 'dependent variables'. The rather crude elementarism of stimulus-response psychology was not affected by this change of terminology, nor was its simplistic view of causality.²

Switching from stimulus-response to variables did have certain tactical advantages. First of all, the historical link with an out-of-date quasi-reflexological schema was finally broken. This gave the language of dependent and independent variables a greater apparent degree of theoretical neutrality than the language of stimuli and responses. It was therefore better suited to become the common language of disciplinary consensus. Different interpretations of what variables represented were permissible, as long as all psychologists agreed that the units of their investigative practice were 'variables'. Secondly, the language of variables could accommodate the practice of psychologists who were engaged in establishing correlations among measures, e.g. personality traits, that had not been experimentally manipulated and hence were not expressible in the language of stimulus and response. The new language was thus broader in scope and more flexible.

The ultimate source of the concept of a variable lies in nineteenth-century mathematics. But how did it find its way into the discipline of Psychology? An important clue is provided by the fact that prior to the 1930s there already existed a special sub-category of psychological literature in which the concept of a variable is present as a matter of course. That special category consists of texts and articles in the area of psychological statistics and the closely related area of educational statistics. In this literature the concept of a variable is employed routinely in the discussion of statistical regression and techniques of correlation. By the 1920s several statistical texts in which the concept appears were not only being used in courses on psychological statistics (e.g. Jones, 1921; Garrett, 1926; Holzinger, 1928) but were also being widely cited in psychological research articles (especially Kelley, 1923). At this time a background in Galtonian statistics was becoming a normal part of training in psychology, and psychological statistics was being

given a more important place in the curriculum. From being taught more or less incidentally within other courses, psychological statistics was now promoted to the status of a topic that required a course of its own. By 1925 a survey by the American Statistical Association established the existence of twelve such courses in Departments of Psychology, a figure which was considered to be an underestimation at the time (Walker, 1929).

The conceptual groundwork for these developments had been laid some years earlier by mathematical statisticians like Karl Pearson and G. Udny Yule, whose work forms the link between nineteenth-century mathematical and twentieth-century psychological statistics. The concept of the variable was central to their contributions. Thus, one finds that the middle section of Yule's enormously influential statistical text is entitled 'The theory of variables' (Yule, 1911), and the term is also to be found in the titles of articles in Pearson's journal *Biometrika* (e.g. Harris, 1909; Pearson, 1910). Charles Spearman, who learned about correlational techniques from the Galton-Pearson School, was probably the first to refer to variables in the psychological literature (Spearman, 1904b). However, at this stage, and for almost three decades thereafter, the term was almost always employed in a strictly statistical context, i.e. in the course of discussions of correlational methods. Only occasionally is the term used in a substantive psychological, rather than a statistical, context (e.g. Knight and Remmers, 1923).

The widespread use of the concept of the variable in a purely psychological context during the 1930s and 1940s must be seen against this historical background. When prominent figures like Tolman and Woodworth began to use this concept in constructing a framework for *all* psychological theorizing and experimenting they could count on it being generally familiar to their by now statistically educated colleagues. That is why they never bother to explain what they mean by this key term, even though it plays such an important role in their exposition. In order to become objects for the type of research psychologists were increasingly engaging in, psychological phenomena like retention, task difficulty, situational knowledge, motivation, affect, etc. did indeed have to be reconstructed as statistical variables. It was perhaps only a matter of time before the originally clear distinction between the psychological kernel and its statistical representation would become blurred.³ But before we trace this process in the empirical literature we must note the remarkable *theoretical* elevation of the category 'variables' that took place during the 1930s.

Variables enter theoretical discourse

The first use of the concept of the variable in a systematic theoretical manner occurs in E.C. Tolman's *Purposive Behavior in Animals and Men*,

published in 1932. There Tolman presents his new version of behaviourism, ushering in the era of neo-behaviourism. In the final section of the book, which he entitles 'This system qua system', he steps back and offers a kind of metatheoretical analysis of his own systematic position, and to do this he uses a metalanguage that enables him to compare his own position with others on his own terms. The concept of the variable plays a key role in this analysis. Quite explicitly, he sets out to compare the variables assumed by his own system with those assumed by other systems (p. 395), the other systems being those of Charles Spearman, E.B. Titchener and the Gestaltists.

What is most remarkable about Tolman's procedure here is that variables are given the status of *psychological* constructs. Previously, variables had been part of the *empirical methodology* of psychologists, at least of those psychologists who followed the Galtonian paradigm. Variables were empirical constructions entailed by the application of correlational techniques to psychological data and nothing more. But now variables were beginning to acquire a psychological, and not merely a statistical, existence.

Previously, whenever psychologists had employed the relevant statistical techniques, they had to give their empirical data a variate form, but this had not at first had much impact on the way in which psychologists formulated their theorizing about their subject matter. The distinction between the level of empirical *statistical* technique and theorizing at the level of underlying *psychological* processes remained relatively clear. Even Spearman and the factor analysts maintained a consistent terminological distinction between the *variables* that entered into empirical correlations and the *factors* that were assumed to underlie these correlations. This left room for fundamental disagreements about the nature of the underlying factors, as the controversy between Spearman and G.H. Thomson illustrates (Thomson, 1920).

However, by the time that Tolman was working on his book this distinction was beginning to wear thin. It is noteworthy that he introduces the crucial chapter in which variables are given a *theoretical* status with a footnote in which he expresses his gratitude to his colleague, R.C. Tryon, 'for stimulating much of the present chapter [and] for a number of the ideas incorporated in it' (Tolman, 1932: 395). Now, Tryon was a specialist in the correlational methods which then dominated psychological statistics. In an article published in the *Psychological Review* three years before the appearance of Tolman's book (Tryon, 1929) he had addressed the question of how the existence of a statistical correlation between two variables was to be interpreted. Two features of his approach to the problem are of interest in the present context. First, in his discussion of underlying factors common to the correlated empirical variables he uses the terms 'common factor' and 'common variable' interchangeably. In other words, he no longer considers it necessary to distinguish between the component of a statistical

technique and a hypothesized psychological entity. Secondly, Tryon emphasizes a *causal* interpretation of purely functional mathematical relationships between variables, though in this respect he was not without predecessors.

These developments prepared the ground for the position adopted by Tolman, where the remaining link of the concept of the variable to specific statistical techniques is dispensed with altogether. Any and all postulated psychological processes are now categorized as 'variables'. In the case of factors derived from individual differences research, like Spearman's 'mental energy' or 'self control', this hardly goes beyond what Tryon had already implied, but when we find Titchener's 'sensations', 'images' and 'affections' referred to as variables we know that we have entered new territory. And when we encounter Tolman's discussion of Gestalt Psychology in terms of variables identified only as 'Gestalts', it becomes clear that variables are no longer what they used to be. Indeed, Tolman invents a new category of what he calls 'intermediate' or 'intervening' variables which function as 'behavior-determinants' (Tolman, 1932: 412). They include innate, genetic, capacities as well as learned capacities and 'purposive and cognitive determinants' acting within particular stimulus-response sequences. Any hypothetical intra-individual event, force, disposition, process, state, ability, etc. that is assigned a causal role in behaviour is now subsumed under the category of 'intervening variable'. Tolman had introduced the key term for a new metalanguage that would facilitate intra-disciplinary discourse across the deep divisions created by differences in subject matter and theoretical preference.

His innovation proved popular and, within one decade, had come to define the terms of the dominant neo-behaviourist discourse (Bergman and Spence, 1941; Hull, 1943a). That made it legitimate to refer to anything that was the object of psychological investigation as a variable, irrespective of its empirical or theoretical status, or of whether it indeed had a variate structure. By the 1950s, the metalanguage of variables had become the common medium for all branches of the discipline aspiring to the honorific designation, 'science'. At that time, the editor of a multi-volume self-presentation of the discipline, sponsored by the American Psychological Association, advised all contributors to describe the structure of their systems in terms of an 'exhaustive itemization of systematic independent, intervening, and dependent variables' (Koch, 1959: 717). His own discussion of the entire range of contributions began with a section on 'The intervening variable paradigm for theory "construction"' (Koch, 1959: 733). Tolman's example had been extraordinarily effective.

In part, this was probably due to the strong link between the language of variables and the already well established language of stimulus and response. Both implied the same strictly linear conception of psychological causality. Woodworth had formulated this conception in the

form of his S - O - R schema, where O represents events within the organism. These are conceptualized as hypothetical or invisible stimulus-response links. In Tolman's version, the place of intra-organismic events was taken by 'intervening variables'. These were thought of as forming the links between independent and dependent variables, much as the O term had linked stimulus and response in the older scheme. In both cases the representation of situations and actions in terms of discrete, logically independent elements became the model for the conceptualization of all psychological reality. Just as behaviour and its situational setting were represented as composed of a multitude of distinct stimuli and responses, or independent and dependent variables, so everything within the individual was now represented in terms of a multitude of intervening variables. Both Woodworth and Tolman aimed at providing the means for what they regarded as a scientific way of describing purposeful activity. Their solutions resembled one another so strongly that Tolman (1932: 423) considered them to be 'siblings'.⁴

The metalanguage of stimulus-response and that of variables also played a similar role sociologically. Both incorporated the principle that the constructive activity of psychologists in their investigative practice was to be regarded as the ultimate meaning conferring operation for all permissible psychological concepts. In the earlier version this was accomplished by insisting on the reduction of all psychological concepts to the elements in terms of which psychologists constructed their experimental investigations. In the later version, the same basic principle soon began to sail under the flag of 'operational definition'. By 1936, Tolman (1936/1951) was referring to his approach as 'operational behaviorism', a term that nicely described much psychological discourse and practice beyond the confines of dogmatic behaviourism. The idea that psychological practice was to define the meaning of psychological concepts was effective, not only because it was an expression of professional privilege, but also because it erected a sharply defined boundary between scientific and lay discourse, especially where the same terms occurred in both. In those large parts of psychology whose vocabulary depended heavily on everyday language the survival of a vigorous scientific community could hardly be guaranteed without this kind of boundary.

Variables in research practice

As we saw earlier, the *technical* term 'variable' became quite well known to research practitioners in the 1920s through their use of statistical methods. Around mid-century, however, there occurs a profound change in the way the term is used in the research literature. If one looks through empirical research reports published at that time, one finds that

the term is becoming a key component in the metalanguage that investigators use to describe, not just an element of technique, but the real objects of their investigation. For example, researchers increasingly claim to be investigating, not just something called 'personality', but entities called 'personality variables'. Similarly, research on emotion is billed as the investigation of 'emotional variables', and the probing of the subjects' past is discussed in terms of 'past history variables'. More generally, both the objects of investigation and the investigative situation are now described in terms of such categories as 'experimental variables', 'social variables' and 'clinical variables'. One encounters a translation of the diverse features, attributes, components and aspects of the investigated world into a uniform metalanguage that allows for only one kind of interrelationship among them, namely, the additive relationship of variables. Natural language description of situations and events had allowed for the representation of a great variety of elements and of their complex interrelationships. It was however inexact and often ambiguous. The language of variables promised to establish scientific clarity and precision by recognizing only one kind of element and only one kind of interrelationship.

If one examines the empirical articles appearing in prominent psychological journals between 1938 and 1958 (Danziger and Dzinis, 1997), one finds that the proportion referring to variables increased substantially during this twenty-year period. By 1958, three-quarters of the articles in both *The Journal of Abnormal and Social Psychology* and the *Journal of Personality* referred to variables, though in 1938 only a quarter of those in the former journal, and one-eighth of those in the latter, had done so. In two journals representing the 'hard' experimental areas of the discipline, (the *Journal of Experimental Psychology* and *The American Journal of Psychology*), the trend was also present, though not as marked. (For the former, the increase was from 33 per cent to 59 per cent; for the latter from 21 per cent to 39 per cent.) Virtually all the papers examined employed statistical methodology, even in 1938. So the rise in reference to 'variables' cannot be accounted for by a rise in the number of studies making use of variables in a technical sense, because virtually all the studies did so. It is not the need to describe a specific technology which accounts for the increasing talk of 'variables', but a new way of describing the reality investigated.

There were obviously factors at work in the research community which favoured increased reference to variables. However, the influence of the new theoretical metalanguage of 'independent', 'dependent' and 'intervening' variables does not seem to have been among these factors, at least not initially. In 1938 not a single empirical paper in any of the four journals employed these terms, in spite of the fact that they were by then in theoretical use. Even in 1948, only six out of a total of 209 empirical papers in four journals refer to 'independent' and 'dependent' variables, and 'intervening variables' appears in exactly one case. By

1958 the situation is beginning to change, though even now only seven out of 390 papers mention 'intervening variables'. However, 27 per cent of the studies reported in *The Journal of Abnormal and Social Psychology*, and 17 per cent of those in the *Journal of Personality*, now refer to 'independent' and 'dependent' variables. In the journals devoted to the more traditionally experimental areas of Psychology this proportion tends to be somewhat lower, reaching 14 per cent in the *Journal of Experimental Psychology* and only 5 per cent in *The American Journal of Psychology*.

By this time several influential texts for research students had made their appearance which had in common a definition of the experimental method in terms of independent and dependent variables (Festinger, 1953; Underwood, 1949; Woodworth and Schlosberg, 1954). The last of these was actually the second edition of a previously mentioned pre-war text (Woodworth, 1938) that had pioneered this way of conceptualizing psychological experiments. In the words of a historian, it 'encouraged future psychologists to think of their research problems and formulate their research questions in terms of one or two independent variables, and not in terms of a multidimensional space of reciprocally interacting factors' (Winston, 1990: 398). By the 1950s this message had been echoed by other textbook writers, and its effects had become clearly noticeable in the empirical literature of the discipline.

Ironically, the message seems to have been heeded a little more faithfully in the 'softer' areas of the discipline than in the traditionally experimental areas. Our count indicates that in 1958 the overall proportion of papers making explicit reference to independent and dependent variables is more than twice as high in the journals catering to research in social, abnormal, and personality psychology than it is in the traditionally experimental journals. This suggests that among the contributors to the journals with a relatively shallow experimental tradition there may have been a few diligent apprentices, anxious to show off their new skills by an invocation of the buzz words of the now fashionable version of experimentalism. Contributors to those journals that had an old and relatively secure experimental tradition behind them would presumably be able to continue in this tradition without the new experimental rhetoric. That would apply particularly to *The American Journal of Psychology*, the oldest of these journals.

One gains a better perspective on this historical pattern if one extends one's horizon beyond the discipline of Psychology. Variables were also becoming popular in the neighbouring discipline of Sociology. In *The American Journal of Sociology*, for example, there was also a marked and steady increase in references to variables over the twenty-year period – from 10 per cent in 1938 to 58 per cent in 1958. Actually, one school of American sociological thought had attributed a kind of higher meaning to variables some years before psychologists did. A leading representative of that school expressed this meaning as follows:

One method of obtaining control in the social science laboratory is, first, to determine the significant factors, or variables, which influence behavior, and then to find out for each its quantitative value in extent or degree. In this way, where it is possible, the social sciences obtain what is an approximation of the controlled experiment in the method of the physical sciences. (Burgess, 1929: 47)

Analysis of situations into quantitative variables was considered to be the way in which the social sciences could successfully imitate their admired model, the physical sciences. It is not unlikely that a significant number of psychologists would have cherished similar sentiments at the time. Among other things, the metalanguage of variables seems to have given expression to a swelling tide of 'physics envy' among American social scientists. If one talked, not of personality, but of personality variables, for example, one was talking like a scientist.

One difference between the situation in Sociology and that in Psychology was that, in the case of the latter, the adoption of the language of variables had a more automatic, non-reflective, quality. Such critical voices as existed (e.g. Cantril, 1950) were either marginalized or simply ignored. In Sociology, however, there was quite a powerful alternative tradition – particularly in the form of symbolic interactionism – which provided a basis for a direct attack on the inappropriate application of the concept of the variable. In 1956 Howard Blumer, a figure of considerable influence in American Sociology, launched a highly visible critique of the use of variables in his discipline. That use, he thought, was too often based on the 'basic fallacy' that independent variables exerted their influence 'automatically' without the intervention of interpretive processes among the persons acted upon. Analysis in terms of variables had become a way of eliminating questions of meaning from the explanation of human conduct. A variable constitutes 'a distinct item with a unitary qualitative make-up' (Blumer, 1956: 688), but the items that are important in the meaningful world of human action are neither clearly distinct from one another, nor do they remain qualitatively unchanged, irrespective of context. Blumer's analysis may actually have strengthened the resolve of some of his colleagues to stick to their variables, for the attraction of this style of social scientific practice was precisely that it enabled one to replace the messiness and ambiguity of the 'subjective' analysis of meaning with the investigator's distinct constructs that always remained the same, no matter what the context.

A more detailed textual analysis of empirical papers published in psychological journals during the period under review indicates that talk about variables did indeed have the function of avoiding the issue of meaning. Explanation in terms of the meaning of situations for experimental subjects is consistently eschewed in favour of a model in which persons 'respond' under the 'influence' of 'variables' that have the solidity of physical objects. A typical example is a study of Rorschach

responses in relation to depression scores on a widely used personality inventory, the MMPI (Blake and Wilson, 1950). Both measures are of course constructions of scientific psychology which depend on the interpretations that individuals make when placed in certain situations. However, when the study is reported in approved journal form, there is no reference to this. Instead, we get talk of depression as an 'adjustmental variable [which] directly influences perceptual selectivity' (p. 459). Referring to depression as an 'adjustmental variable' ('operationally defined' by MMPI scores) transforms it from a qualitative feature of subjective worlds into an objective entity that varies only in degree and that has causal effects on other objective entities, like 'Rorschach determinants' in our illustrative example. The language of variables was thus the perfect vehicle for an eclectic kind of neo-behaviourism that wished to banish subjective meaning as an explanatory principle without any commitment to *specific* mechanistic alternatives.

All that remained of the mechanistic fantasies, so dear to the card-carrying behaviourists, was a *general* commitment to the explanation of human conduct in terms of the causal efficacy of 'operationally defined' (but objectively existing) variables, additively interacting. Two inter-related conceptual moves were necessary in order to accomplish the construction of this position. Both depended on fudging the existential status of the variables employed in psychological research. In the empirical literature of the time we find a widespread tendency to slide between the use of 'variable' as a label for a statistical construct and as a description of an objective natural force with causal efficacy. Thus, the variable 'depressive tendency' in our illustrative article refers both to MMPI scores and to something that has a distinct, causally active, existence in human individuals. Now, about the existence of MMPI scores as variables there can be no doubt, because these are artefacts that we have constructed along the lines prescribed by a particular statistical model. But in themselves such variables are not very interesting. What makes them psychologically interesting is the imputation that they are identical, or at least closely related, to other variables that operate out there in the real world without our intervention. The step from variables in the modest statistical sense to variables in the grand ontological sense is a huge one, yet, in writing up their research, psychological investigators were constantly taking it, apparently without a moment's reflection.

This is not to say that, in the psychological literature of the time, one cannot find examples of careful distinctions between hypothetical causal determinants and their empirical referents, but these remained largely limited to theoretical texts. Very few empirical research papers bothered with such niceties, and, judging by the published products, neither did journal editors or reviewers. In transposing the category of 'variables' from a statistical to an ontological context psychologists had committed themselves to a nebulous language whose ambiguities often proved

convenient. Empirical reports usually did not limit themselves to modest statements about the contribution of specific variables to statistical variance but talked in terms of the 'influence', the 'direct influence', the 'effect', of particular variables. In the published journal texts the variables that investigators had constructed by means of their measuring instruments often appear as causal agents, variously described as 'determinative factors', 'influencing variables', 'determining variables', which 'affect' psychological processes, 'produce' effects, and play a 'determinative' role. This could just be taken as a loose way of referring to observed statistical relationships, but the context makes it quite clear that ontological claims were being made, and that the reader was being invited to think in terms of causal forces at work in real individuals.

The evidence of the journal literature suggests that this aspect of the language of variables was developing in a practical research context without any explicit reference to the contemporaneous theoretical discourse about 'intervening variables'. Yet there was a remarkable symmetry between the two. The theoretical discourse of neo-behaviourism, focusing on the status of hypothetical entities, tended to reduce them to observed relationships among empirical variables (sometimes adding a meaning that was characteristically designated as 'surplus'). At the same time, research investigators, who were constructing these empirical variables in their daily practice, were endowing them with theoretical meaning by using a language which blurred the distinction between hypothetical and empirical elements. The starting points and the directions of theoretical and empirical discourse were different, but the products were very similar. Theoretical discourse started with theoretical terms and moved in the direction of giving them the strongest possible link to empirical variables. Empirical discourse started with the latter and moved in the direction of equating them with their theoretical counterparts. The two levels of discourse converged in producing a version of psychological reality that amounted to a reification of psychological research practice.

Variables as lingua franca

Earlier in this chapter it was suggested that the language of stimulus-response provided a kind of lingua franca which made it possible for experimental psychologists to communicate with each other in spite of the theoretical differences that existed between them. There is evidence that the language of variables functioned in a similar way for psychologists in the areas of clinical and personality research.

We have already seen that, contrary to what one might have expected, it was not the journals covering the 'harder', more experimental, parts of the discipline that led the way in adopting the language of variables. In the post-World War II period it was first of all the *Journal of Personality*

and then also *The Journal of Abnormal and Social Psychology* that showed a massive increase in the proportion of articles referring to variables. The meteoric rise of the 'variable' was first signalled by the *Journal of Personality*, which has the lowest proportion of articles using the term in 1938 but by far the highest proportion in 1948. Ten years later, three-quarters of its articles were employing the term. Moreover, when one examines the textual context of 'variable', one finds that the commonest pairing in *all* the journals is with 'personality'. During this twenty-year period, the term 'personality variables' changes from a rarely used form to a ubiquitous phrase in the psychological literature. What was behind this surge?

In 1938 the area of personality psychology had just entered a phase of consolidation in which a common model of empirical research provided a link between previously incommensurable theoretical orientations. Until the late 1930s personality psychology had been essentially an arena for competing schools, but at that point the three virtually simultaneous texts by Gordon Allport, Ross Stagner, and Henry Murray established an eclectic position that privileged empirical research over theoretical principles and thus founded the field as it still exists today. Although the Allport text came to enjoy a certain didactic pre-eminence, it was the Murray text that appears to have been the most significant from the point of view of day to day research practice. It was certainly the one most likely to be cited in research papers.

For the fate of the 'variable' in personality psychology Murray's *Explorations in Personality* was of prophetic significance. Published in 1938, at a time when the use of the term 'variable' in the relevant journal literature can only be described as sporadic, this comprehensive research monograph was remarkable for its ubiquitous use of the term. Indeed, it contains a hundred-page chapter entitled 'Variables of personality'. It is hardly surprising that it is frequently cited in journal articles that participate in the rapidly expanding usage of the term during the next decade. But what accounts for the sudden prominence of 'personality variables' in this 1938 text? It certainly had nothing to do with its author's own theoretical predilections. Henry Murray was perhaps the last person one would have expected to encourage the use of such terminology. His background was in Jungian psychology, and he was regarded with great suspicion by his more conventional, scientifically inclined colleagues in the Harvard Psychology Department (Triplet, 1992).

However, the most striking feature of *Explorations in Personality* is that it was a collaborative product rather than the expression of the views of a single author. Although Henry Murray was its chief architect, a dozen psychologists and another dozen graduate research assistants are listed as collaborators. These were all individuals who participated in the research of the Harvard Psychological Clinic, generously funded by the Rockefeller Foundation. They were quite a motley crew, described by

Henry Murray as 'a group composed of poets, physicists, sociologists, anthropologists, criminologists, physicians; of democrats, fascists, communists, anarchists; of Jews, Protestants, Agnostics, Atheists; of pluralists, monists, solipsists; of Freudians, Jungians, Rankians, Adlerians, Lewinians, and Allportians' (Murray, 1938: xi).

Although their spokesman bravely declared that they 'did not wish to succumb to the great American compulsion to co-operate' (*ibid.*), co-operate they did. The relative coherence of their 700-page research monograph is evidence of that. How did they achieve such a feat? By building on the one thing they had in common: some measure of faith in the appropriateness of the empirical techniques then in vogue for providing significant knowledge about human individuals. These techniques included the familiar statistical methods for estimating the degree of relationship among variables. This meant that whatever their theoretical and ideological differences they agreed that in practice the things they were trying to investigate had to be treated as variables. And with that agreement they had provided themselves with an escape from their Tower of Babel. They had found an apparently neutral language that made a modicum of communication possible, a language that actually enabled them to formulate a common research objective in spite of the profound disagreements that otherwise divided them. Murray, their spokesman, states this objective very succinctly when introducing the report on their labours: 'What are the fundamental variables in terms of which a personality may be comprehensively and adequately described?' (*ibid.*: x). The participants in this collaborative research endeavour could continue to hold whatever pet theories they liked, as long as they agreed to define their research objectives in the language of variables. This was the basis on which personality psychology could surmount its fractionation into feuding schools and construct something that came to resemble what Thomas Kuhn would later characterize as 'normal science'.

Using the language of variables at the Harvard Clinic also proved convenient for another, related, reason. As so often happens in a clinical institutional context, there were the conflicting pulls of practical clinical work and research work to contend with. The Clinic's 'diagnostic council', a group that was at the core of its collaborative endeavour, had to try to reconcile these demands. Not surprisingly, theoretical development was an early casualty of the compromises that had to be made: 'Time constraints meant that the diagnostic council decisions were almost always based on statistical averages of ratings rather than on discussion, and the whole enterprise was oriented toward achieving practical results rather than developing theory' (McLeod, 1992: 10). In such a situation, the language of variables would provide an apparently innocent, yet convenient, medium for achieving a culturally sanctioned compromise. The elaboration of 'personality variables' and their inter-correlations would provide an actuarial basis for practical diagnostic

decisions and also a way of conducting quantitative studies that qualified as research. Discussion of the meaning of such variables would even provide an acceptable substitute for theorizing about hypothetical structures and processes in the old style inspired by psychoanalysis.

For psychologists faced with problems similar to those of the Harvard group the new language of variables had obvious attractions. Just as this language had provided the basis for a necessary minimum of understanding among the members of the Harvard group it came to provide the common language in which personality psychologists beyond this group were able to communicate with each other in spite of their often considerable theoretical differences. Over a longer period of time this arrangement had the effect of making these theoretical differences seem less important. Eventually, the old theories became irrelevant in a research context and were replaced by other kinds of theory that were essentially a reflection of methodological and practical commitments.

The situation that led to the widespread adoption of the language of variables at the Harvard Psychological Clinic, and then in personality psychology generally, was essentially duplicated in other parts of the discipline, notably in abnormal and social psychology. In the 1940s these areas were hardly less marked by deep theoretical and ideological divisions than the area of personality psychology. One need only think of the profound ideological conflicts that received their expression in the formation and early days of the Society for the Psychological Study of Social Issues and the innumerable schools of thought about how best to explain psychological disorder. Here too, the language of variables made it possible at least to communicate on one level and thereby to satisfy an important condition for the formation of something like a scientific community. At the same time, it provided a discursive framework for a kind of research that was compatible with an institutional practice based on the normative use of interpersonal variance for diagnostic purposes. Many individuals who were active in the area of personality psychology were also involved in either abnormal or social psychology, so the spontaneous transfer of such a convenient medium from one area to another would have been relatively easy.

Theoretical reflections

It is time for some critical reflections on the implications of the historical analysis presented in this chapter. It has been suggested that psychological investigators were ready to accept their own quantitative research technology as the one philosophically unproblematic component of their discipline. Whatever suspicion there might be of mentalistic concepts, and however inconclusive the theoretical debates among the psychological schools, a common statistical research technology would form the solid rock on which a scientific psychology could be built.

No doubt, the language of variables proved so seductive at the time because it was tied to apparently neutral statistical techniques that a great many psychologists were using in any case. And this apparent neutrality extended to the term 'variable', even when it was no longer used in a strictly statistical context. But of course the neutrality was only apparent, because the description of personality and of social situations as constituted by sets of distinct variables was in fact theory laden. Recognition of this was hindered by the seldom questioned assumption that the step from methods of investigation to independently existing objects of investigation could be executed quite unproblematically. Yet, in the war against metaphysics the innocent-seeming technology of analysing variables played the role of a Trojan Horse. For if one proceeded as though such an analysis provided a *direct reflection* of psychological reality one was, in effect, making a metaphysical choice. One was now committed to a particular model of psychological reality as composed of a certain kind of element and a certain kind of relationship. In this model there could only be elements that retained their identity irrespective of the relationships into which they entered, all elements being logically independent. Such elements would vary only quantitatively, not qualitatively, between individuals and between situations, and the relationship between them would be essentially additive (cf. Runkel, 1990: ch. 8).

Such a model certainly had its uses for purposes of making practical predictions within a limiting framework of conditions. If one's goals were essentially economic, and one had no mandate or interest in changing existing social arrangements or values, an analysis in terms of variables would often be perfectly adequate. Thus, if one accepted a particular criterion of success and failure (prescribed by prevailing values), one could calculate the relative efficiency of investing fixed resources in one direction rather than another. For instance, a given overall level of educational success can be more efficiently achieved by investing in the education of individuals with high rather than low scores on certain tests. For this purpose it is unnecessary to develop any kind of psychological theory about the nature of mental ability – an analysis of statistical relationships among empirical variables is all that is required. Of course, if social criteria of success or the distribution of social resources were to be questioned, some sort of psychological theory would probably be invoked, both by the advocates of change and by the defenders of the status quo. At that point, psychological engineering would no longer provide an adequate model.

In promoting the language of variables to the status of a metalanguage, psychologists had adopted the language of psychological engineering as a universal medium for theoretical exposition. In other words, they had conflated conceptual control and instrumental control. Philosophers of science have pointed out that scientists 'may achieve remarkable instrumental control without correctly understanding their

instruments' (Nickles, 1989: 311; also Hacking, 1983), a situation common in the field of engineering. Theoretical understanding and the technical production of particular effects are quite different in principle, though historically various links between them may develop (Maricas, 1987: ch. 13). In the present case, the prescriptive equation of conceptual control with instrumental control imposed severe limitations on what was admissible as psychological theory. Causality was reduced to a crude concatenation of antecedents and consequents, while complex systems and meaningful patterns were replaced by statistical relationships among a never ending list of elements that always remained logically independent of one another. Of course, the restrictions imposed by this language were broken on numerous occasions, not least by Tolman himself, and then variables became mere *indices* or signs that pointed to a different kind of reality. But such moves were generally made apologetically, because they were perceived as running counter to the norms of real science. By the 1950s the language of variables had taken over the scientific high ground. Those who limited themselves to it never had to answer for themselves, while those who went beyond it were obliged to offer excuses.

Ironically, the way in which variable research was deployed in social scientific practice constituted one of the fundamental differences between it and the common practices of physical science (Winston and Blais, 1996). In the social sciences the command to look for variables was taken to mean that phenomena ought to be investigated in terms of their variation – in Psychology, typically, inter-individual variation. Had physicists proceeded like this they would have attempted to study the *differences* among a variety of falling bodies, or the differences among a variety of warm bodies. Fortunately, this was not their approach, for had it been, it is highly improbable that they would ever have arrived at the law of gravity or the principles of thermodynamics. At least in the modern period, it was clearly understood that the explanation of physical phenomena was one thing and the explanation of their variation quite another. Had it not been for the lack of this basic insight in large segments of the social sciences, it is unlikely that the analysis of variation would have achieved the pre-eminent position it did.⁵

The present chapter has been devoted to an exploration of the ways in which the disciplinary language of Psychology was shaped by the nature of its investigative practice, and more particularly, by the way in which that practice was understood within the discipline. What emerged historically was a kind of isomorphism between the structure of investigative practice and the structure of the objects of investigation. Before they could become objects for scientific psychology, terms referring to psychological objects had to be translated into the terms that were appropriate within the framework of psychological research. Thus, social action, for example, had to be translated into responses to social stimuli, and personality had to be translated into personality variables.

This replaced categories taken from everyday discourse with new categories that were based on a reification of the terms imposed by particular forms of investigative practice.

This historical process represents only one aspect of the complex and intimate relationships that developed between psychological practice and psychological theory. Another aspect, analysed by Gigerenzer (1991), involves the function of methodological tools, e.g. inferential statistics, as sources of theoretical metaphors and concepts. It is becoming more and more apparent that the technology of psychological research – once regarded as a neutral and passive tool – has played, and continues to play, a vitally important determining role in the history of the discipline and of the discursive objects it constructs. There is a whole family of interrelationships between concepts and practices that has not yet been fully explored or even named. In the meantime, perhaps one should update what William James famously referred to as 'The Psychologist's Fallacy'. He defined this as 'the confusion of his own standpoint with that of the mental fact about which he is making his report' (James, 1890, I: 196). In the twentieth century one would want to substitute the more noncommittal 'psychological' for 'mental'; but above all, one would want to be clear about the fact that psychologists' 'own standpoint' has been decisively shaped by their methodological commitments. These commitments have been a mixed blessing at best. On the one hand, they have been a source of productive heuristics and have allowed something like Kuhn's 'normal science' to go ahead. On the other hand, they have blinkered the discipline conceptually and taken it down some very long blind alleys.

Notes

1. William James, for instance, employs the term once in his *Principles of Psychology* (1890, I: 59) and again in a discussion of Dewey's ideas published in 1904. (I am indebted to Paul Ballantyne for finding the relevant passages.)

2. That view can be summarized in terms of the following assumptions, common to the stimulus-response scheme and analysis in terms of variables: '1. Cause comes before effect. As a corollary, cause and effect are separate, non-overlapping events. 2. Cause and effect appear in local, distinguishable episodes having a beginning and an ending. . . . As a corollary, the beginning and the end are recognizable. 3. The laws of nature are everywhere the same . . . if a law holds at one place or time, it will hold in every place and time' (Runkel, 1990: 78).

3. Jerome Kagan (1989: 130), for example, seems to be suggesting this when he writes: 'During this century the disciplines that study behavior, thought, and affect have favored continua over categories and linear over non-linear relations. These presuppositions . . . dominate the modern psychological laboratory, partly as a consequence of the dissemination of statistical procedures in the interval between the two World Wars. . . . Investigators who ignored statistical analyses were regarded as less competent.'

4. My focus here (as elsewhere) is purely on Tolman as a voice in the public discourse of a discipline, not on the intellectual biography of an individual. From the latter point of view, there was clearly a considerable distance between where Tolman started and where

he ended up. He began his career in the mixed empiricist-phenomenological tradition of James, Holt and Perry, but, as Still (1987) has suggested, his absorption in the language and the experimental practices of Psychology pointed him in a different direction. He may have started with insights into the direct perception of purpose, but, 'given his audience, there was an inevitability in the progression of his thought' so that 'he was eventually obliged to express them [his insights] in a language akin to that of the more mechanistic psychology that became dominant during the 1930s' (Still, 1987: 181).

5. Kurt Lewin and B.F. Skinner were two major figures who did recognize the importance of this distinction quite explicitly. Many other European psychologists, e.g. Sigmund Freud and Jean Piaget, recognized it implicitly, which is why they had no use for 'variables'. On Lewin, see Danziger (1992a); more generally, see Lewontin (1985), and Lieberman (1985).

THE NATURE OF PSYCHOLOGICAL KINDS

Historical roots of psychological categories

Traditional prescriptions for the practice of a scientific psychology seem to imply an image of the psychologist as a sort of visitor from outer space (Ingleby, 1990), unaffected by local cultural traditions or the preconceptions embedded in human language. Right at the beginning of this book that image was rejected in favour of one that depicts psychology as a cultural construction with specific historical roots. The body of the book was devoted to an exploration of some of these roots. In light of this exploration, what can we say about the sources of psychological categories? Obviously, these categories all depend on certain semantic traditions and a historically emerging usage. Is it possible to go beyond such general statements by identifying some of the major factors in the historical formation of psychological categories?

I propose to answer this question positively by first recalling the various layers of discourse that have been examined in the preceding pages. At each of these levels a particular set of conceptual distinctions emerges so as to form a specific network of interconnected categories that enables a certain kind of understanding of human subjects.

At a relatively deep level there are broad distinctions that have pervasive effects. In this book we have concentrated on the distinctions introduced by a certain kind of moral discourse that first became fully articulate in eighteenth-century Britain. Aiding both self-understanding and self-justification under new social and economic circumstances, this discourse was based on a fundamental sense of separation between human individuals as well as between individual agents and their actions. In Chapter 3 we saw how this manifested itself in the emergence of a network of categories with modern meanings, such as self, consciousness and motive. The new moral sensibility also involved revolutionary changes in the understanding of ancient categories like reason and passion without which modern conceptions of intelligence and emotion would have remained literally unthinkable.

Had there been no further developments psychology would have remained simply a part of moral discourse. In part, it has indeed